

Day 113 Project

Hybrid Volatility Forecasting for Option Pricing

GARCH vs Heston

1 Introduction

Volatility plays a central role in option pricing. Different models produce different volatility estimates, leading to different valuations.

This project compares:

- GARCH-based volatility (discrete-time)
- Heston stochastic volatility (continuous-time)

2 Framework

We price a European option using:

- Black-Scholes with GARCH forecasted volatility
- Monte Carlo simulation under Heston dynamics

3 GARCH-Based Volatility

Returns are modeled as:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Effective volatility:

$$\sigma_{eff}^2 = \frac{1}{T} \sum_{i=1}^T \hat{\sigma}_{t+i}^2$$

4 Black-Scholes Pricing

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

5 Heston Model

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}$$

$$dv_t = \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^{(2)}$$

6 Monte Carlo Pricing

$$C = e^{-rT} \mathbb{E}[\max(S_T - K, 0)]$$

7 Key Comparison

- GARCH: data-driven volatility forecasting
- Heston: stochastic volatility dynamics
- Pricing differences reflect volatility modeling assumptions

8 Insights

- GARCH adapts quickly to recent market shocks
- Heston captures long-term structural behavior
- Differences in pricing highlight model risk

9 Conclusion

Volatility modeling choice has a direct impact on option pricing. Combining statistical and stochastic approaches provides a more comprehensive framework.